Mechanics Year 2 · Chapter 4

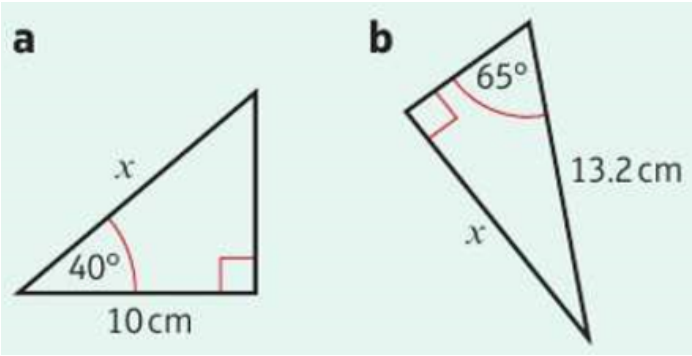
Moments

Turning effects · Equilibrium of rigid bodies · Centres of mass · Tilting

Use $g = 9.8\,\text{m}\,\text{s}^{-2}$ unless stated otherwise. Diagrams are schematic.

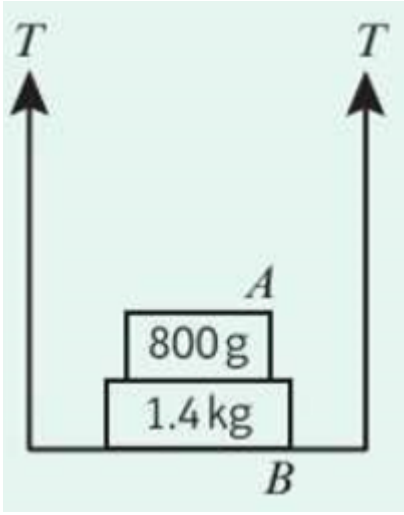
Starter 2

1. Find x in each right-angled triangle.



1a–b

2. The light scale-pan is supported by two strings, each of tension T. Find T, the reaction of B on A, and the reaction of the pan on B.



2

The door problem 3

Why is a door handle usually far from the hinge?



Turning effect

If the perpendicular distance is doubled, what happens to the force required to produce the same turning effect?

Double the distance

Rigid Bodies and Overview 4

From particles to rigid bodies

A particle model ignores an object's dimensions. A rigid-body model includes its size and shape, so it can describe rotation.

A rod is a thin rigid body. A lamina is a thin, flat body.

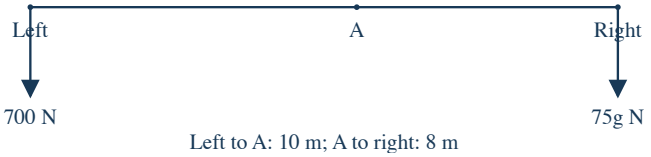
Three ideas in this chapter

1. Equilibrium: the resultant force and resultant moment are both zero.
2. Centre of mass: a uniform rod's weight acts at its midpoint; a non-uniform rod's need not.
3. Tilting: a support about to lose contact exerts no reaction. Use the remaining pivot to take moments.

Moment of a force about a point

$M = Fd$

Here d is the **perpendicular distance** from the point to the force's line of action. Units: newton metres, N m.



A light seesaw carries a 700 N weight on the left and a 75 kg mass on the right. Which way does it turn about A?

Moments

Direction

Quickfire Examples 6

Find each moment about P, including its direction.

1. A 20 N force acts downwards 10 m to the right of P.
2. A 20 N force has a line of action through P.
3. A 10 N force acts at 60° below a horizontal rod, 5 m to the right of P.
4. A 10 N force acts at 30° above a horizontal rod, 2 m to the right of P.
5. A 5 N force acts 65° from the upward vertical, 3 m to the right of P.

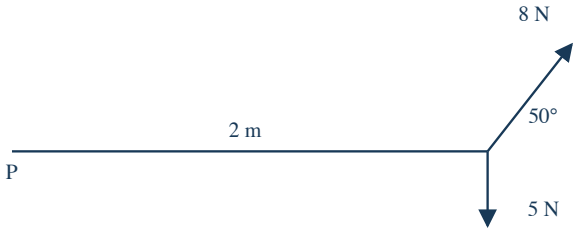
1–3

4–5

For a right triangle with hypotenuse r: opposite side $r \sin\theta$, adjacent side $r \cos\theta$.

Test Your Understanding 7

Two forces act on a lamina. The 5 N force acts downwards, 2 m horizontally to the right of P. The 8 N force acts at $50°$ above a horizontal radius of length 2 m to the right of P. Find each moment.

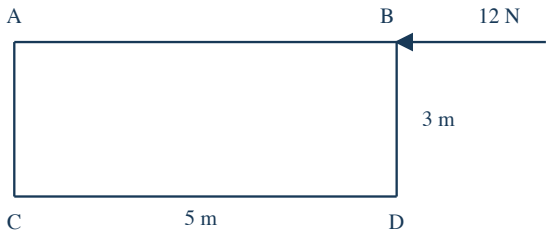


5 N

8 N

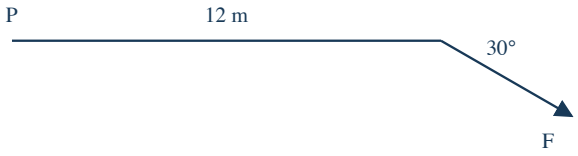
Exercise 4A Page 72 8

4. A rectangle has $AB = CD = 5$ m and $AC = BD = 3$ m. A horizontal 12 N force acts to the left at B. Find its moment about a) A, b) B, c) C, d) D.



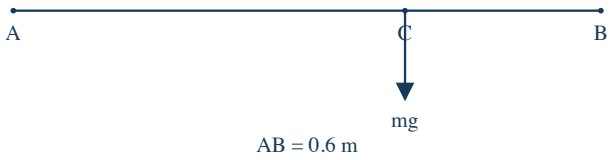
4a–d

5. A force F acts at $30°$ below a rod, 12 m to the right of pivot P. Its moment is 15 N m clockwise. Find F.



5

A mass hangs at C on a horizontal beam AB of length 0.6 m. The magnitude of its moment about A is twice that about B. Find AC.



Working

Resultant moments 11

Resultant moments

Choose clockwise or anticlockwise as positive and add signed moments. Find these resultants about P.

1. 20 N downwards, 10 m right; 10 N downwards, 15 m left.
2. 5 N downwards, 3 m right; 4 N upwards, 1 m right; 3 N downwards, 1 m left.

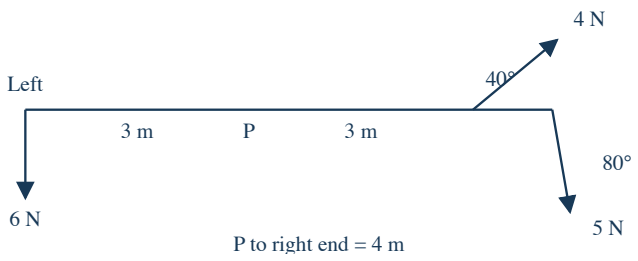
1–2

3. A light rod carries 50 kg at 1 m left of P, and 30 kg at 3 m right of P.
4. A light rod has length x m. Pivot P is 4 m from its left end; a 30 kg mass hangs at its right end $(x > 4)$.

3–4

Angled examples 12

A light horizontal rod has pivot P. A 6 N downward force acts 3 m left of P; a 4 N force acts 3 m right, at $40°$ above the rod; a 5 N force acts 4 m right, at $80°$ below the rod. Find the resultant moment.

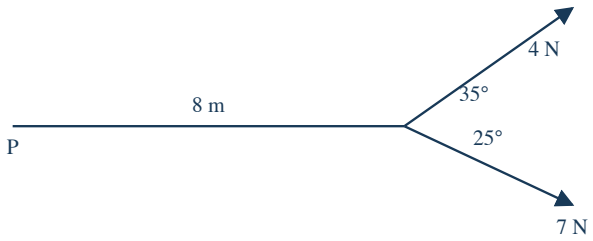


Distances

Resultant

Test Your Understanding 13

Two forces act at a point 8 m horizontally to the right of P on a lamina: 7 N at 25° below the horizontal and 4 N at 35° above it. Find the resultant moment.

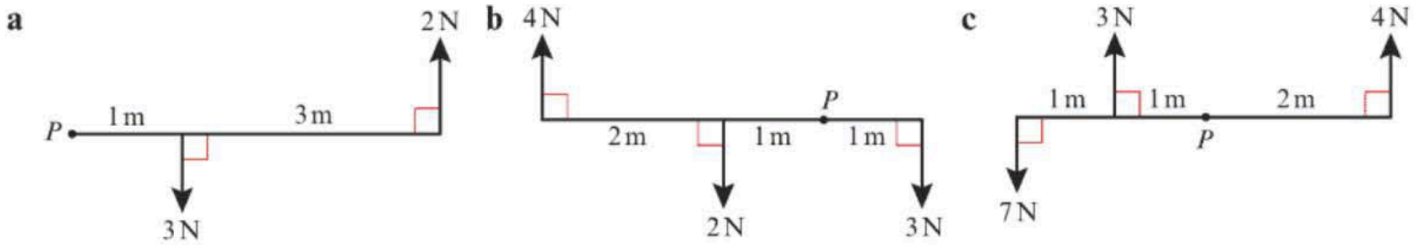


Distances

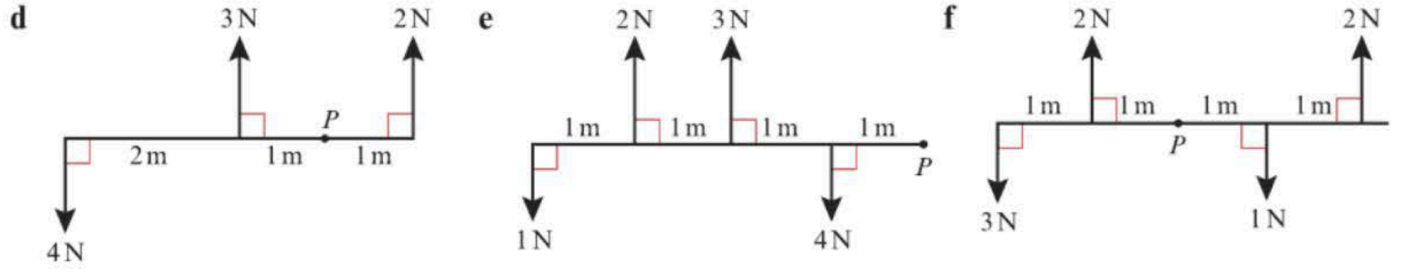
Resultant

Exercise 4B Page 74-76 14

1. Find the resultant moment about P for each light rod.

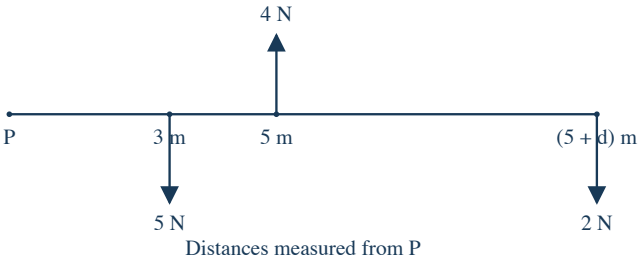


a–c



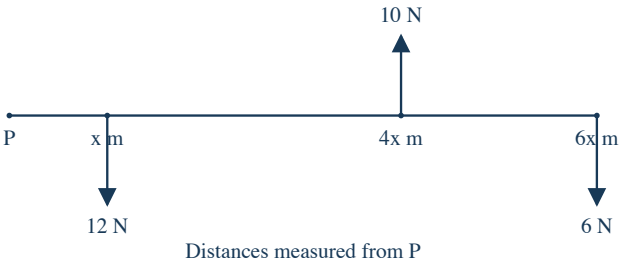
d–f

3. Forces act on a light rod as shown. The resultant moment about P is $17\,\text{N m}$ clockwise. Find d.



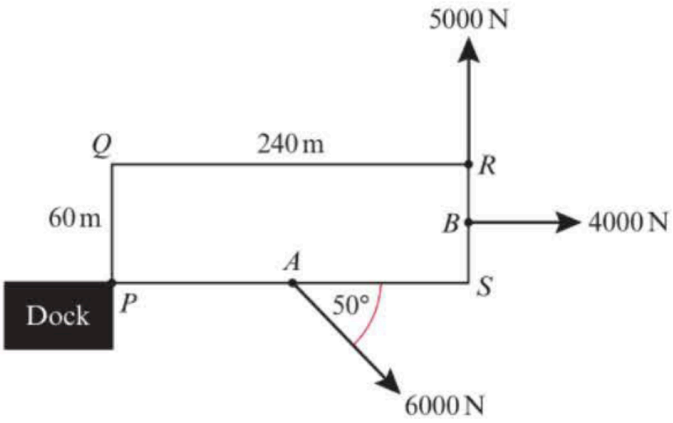
3

4. The resultant moment about P is $12.8\,\text{N m}$ clockwise. Find x.



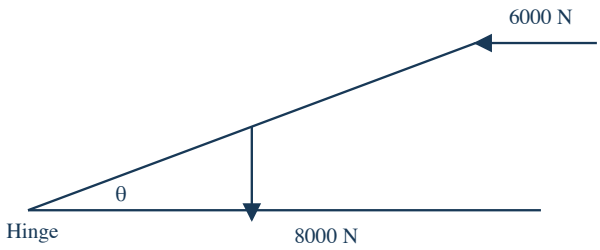
4

5. A cruise ship is modelled as a rectangular lamina $PQRS$, fixed at P. Three tugs exert the forces shown. A is the midpoint of PS, and B the midpoint of RS. Find the resultant moment about P and its direction.



5

6. A uniform drawbridge of weight 8000 N is inclined at θ above the horizontal. A horizontal 6000 N force at its upper end acts towards the hinge to raise it. Show that raising requires $\tan\theta > 2/3$.



6

Extension 19

1. A rectangular barge is 120 m long and 20 m wide. It is fixed at its bottom-right corner P. At the top-left, a 2000 N force acts down-left at 45° to the vertical; at the bottom-left, 1100 N acts down-right at 60° below the horizontal. Find the resultant moment and state a modelling assumption.

1

2. A uniform aircraft door AB, length 4 m and weight 2500 N, is hinged at B. It is inclined at 45° to the horizontal. A vertical upward force at A tends to lift the door. Find the force needed to start lifting.

2

This whole chapter in a nutshell. . . 20

Equilibrium of a rigid body

Both conditions are required:

$$\sum \mathbf{F} = \mathbf{0}, \qquad \sum M = 0.$$

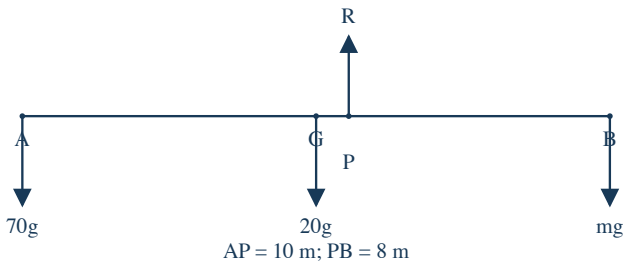
For vertical forces, total upward force equals total downward force. About any point, total clockwise moment equals total anticlockwise moment.

A reliable method

1. Draw the external forces.
2. Place a uniform rod's weight at its midpoint.
3. Resolve forces.
4. Take moments about a point that eliminates an unwanted force.
5. Check units and the physical meaning of the answer.

Example 21

A uniform seesaw has length 18 m and mass 20 kg. Its pivot P is 10 m from the left end. Woody (70 kg) sits at the left end; Buzz (m kg) sits at the right. Find m and the pivot reaction R.

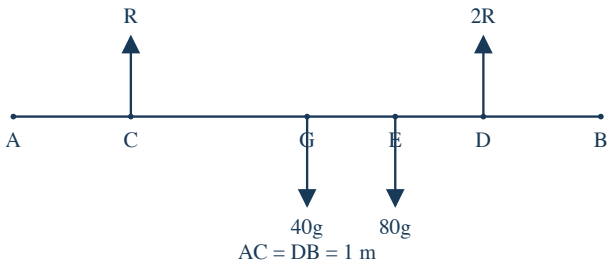


Method 1

Method 2

Two Pivots 22

A uniform beam AB, length 5 m and mass 40 kg, rests on supports C, D with $AC = DB = 1$ m. An 80 kg man stands at E. The reaction at D is twice that at C. Find AE.



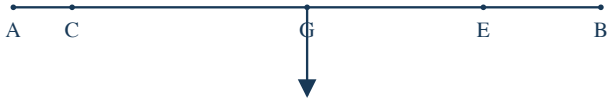
Working

Test Your Understanding 23

Edexcel M1 May 2013(R), Q8. A uniform beam AB is 2 m long and has mass 50 kg. Supports are at C, D, where $AC = 0.2$ m and $DB = x$.

a) If $R_D = 2R_C$, find x.

b) Move D to E, where $EB = 0.4$ m. Put a particle of mass m at B. If $R_E = 4R_C$, find m.



50g

Part b: AC = 0.2 m; EB = 0.4 m

a

b

Exercise 4C Page 78-80 24

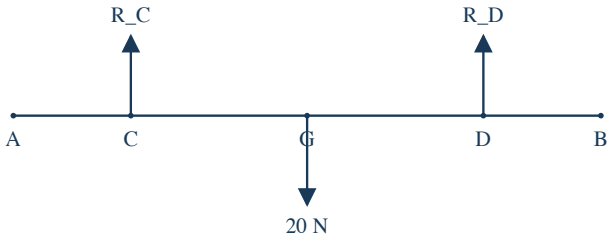
1. A uniform rod AB is 5 m long and weighs 20 N. It rests horizontally at supports C, D. Find the reactions in each case. All distances are measured from A.

a) $AC = 1$ m, $AD = 4$ m.

b) $AC = 2$ m, $AD = 4$ m.

c) $AC = 1.5$ m, $AD = 4$ m.

d) $AC = 1.5$ m, $AD = 4.2$ m.

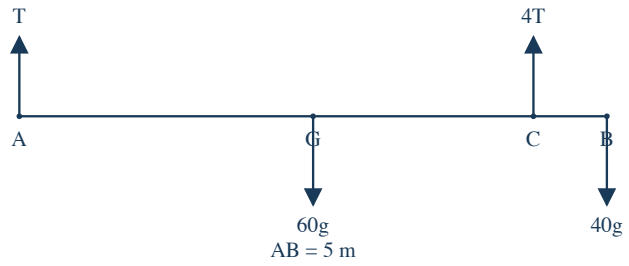


20 N

Schematic: support positions vary

a–d

7. A uniform beam AB, length 5 m and mass 60 kg, carries a 40 kg load at B. Vertical wires at A, C hold it horizontally. The tension at C is four times that at A. Find a) the tension at C, b) CB.

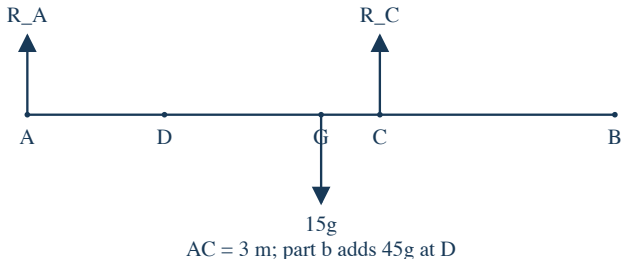


7a–b

8. A uniform plank AB, length 5 m and mass 15 kg, rests at A, C, where $BC = 2$ m.

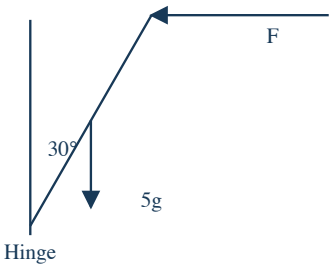
a) Find the reaction at C.

b) A 45 kg person stands at D, making the two reactions equal. Find AD.



8a–b

10. A uniform lever has mass 5 kg and length 1.5 m. It is smoothly hinged to a vertical wall and is held 30° from the vertical by a horizontal force F at its other end. Find F.

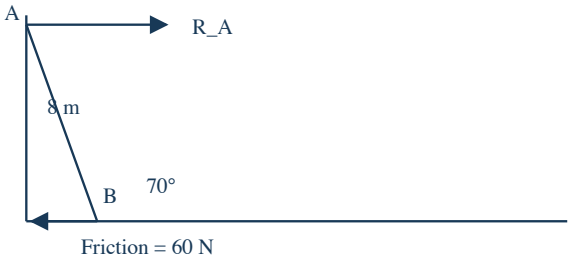


10

11. A uniform ladder AB, length 8 m, leans against a smooth vertical wall at 70° to rough horizontal ground. Friction at B has magnitude 60 N.

a) Find the wall reaction at A.

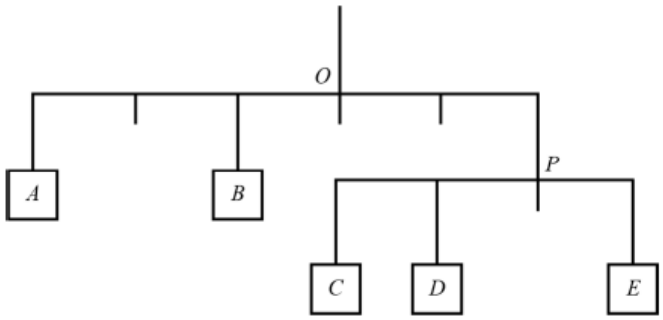
b) Find the ladder's mass.



11a–b

Extension 29

Arrange masses 1, 2, 3, 4 and 5 kg on the five marked positions of this kinetic sculpture. Use each once so both light rods hang horizontally. Marked intervals on each rod are equal.



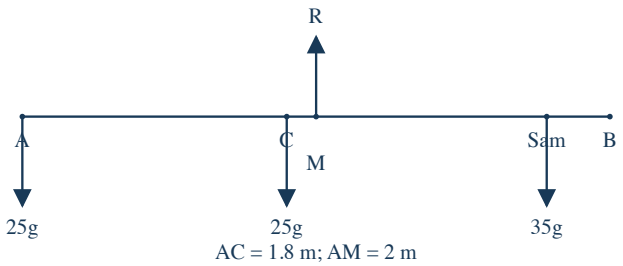
Hints

Arrangement

Centres of Mass 30

A uniform rod's centre of mass is its midpoint. For a non-uniform rod it need not be.

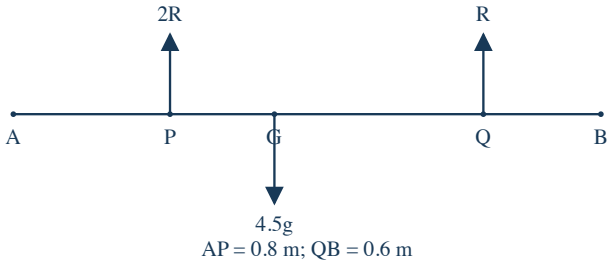
A non-uniform plank AB, length 4 m and mass 25 kg, is pivoted at its midpoint M. Its centre of mass C has $AC = 1.8$ m. Tamsin (25 kg) sits at A. Where must Sam (35 kg) sit to balance it?



Working

Test Your Understanding 31

Edexcel M1 May 2012, Q2. A non-uniform rod AB has length 3 m and mass 4.5 kg. It rests at P, Q, with $AP = 0.8$ m and $QB = 0.6$ m. The reaction at P is twice that at Q. Find a) the reaction at Q, b) the distance AG to the centre of mass.



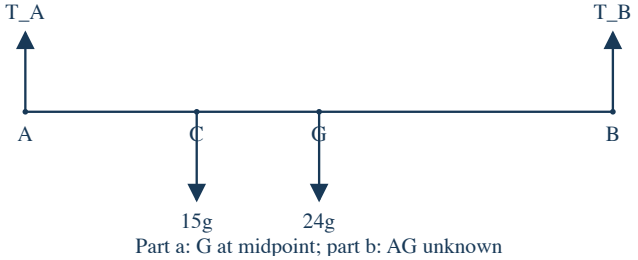
a–b

Exercise 4D Page 81-83 32

5. A plank AB, mass 24 kg and length 4.8 m, carries a 15 kg load at C, where $AC = 1.4$ m. Vertical ropes at A, B hold it horizontally.

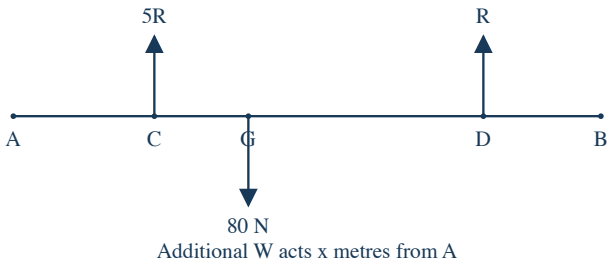
a) Model the plank as uniform. Find the tension at B.

b) In a non-uniform model, the tension at A is 25 N greater than at B. Find the distance of its centre of mass from A.



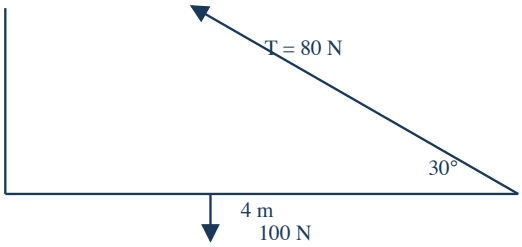
5a–b

7. A non-uniform rod AB has length 25 m and weight 80 N. Supports C, D are 6 m and 20 m from A; its centre of mass is 10 m from A. A weight W is attached x metres from A. If $R_C = 5R_D$, show that $W = 400/(25 - 3x)$.



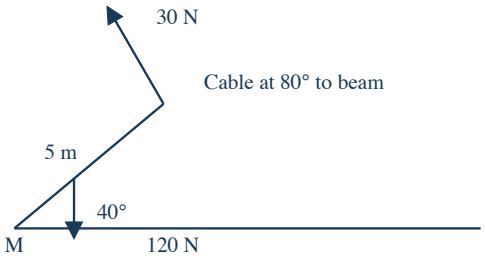
7

8. A non-uniform rod, length 4 m and weight 100 N, is hinged to a vertical wall. A cable at its free end makes $30°$ with the horizontal rod and has tension 80 N. Find the distance of the centre of mass from the wall.



8

Challenge. A non-uniform beam has weight 120 N and length 5 m. It is pivoted at M, inclined at 40° to the horizontal, and held by a cable at its upper end. Cable tension is 30 N at 80° to the beam. Find the distance of its centre of mass from M.



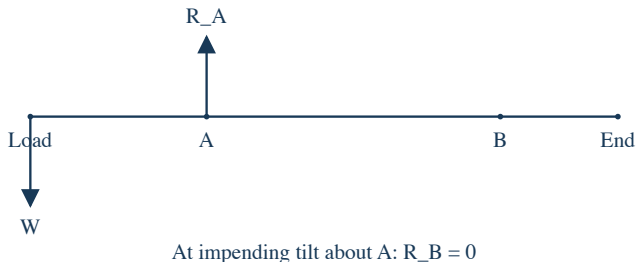
Working

Tilting 36

On the point of tilting

As a load moves beyond a support, the reaction at the other support decreases. At the instant contact is lost, that reaction is zero.

A cable that becomes slack likewise has zero tension.

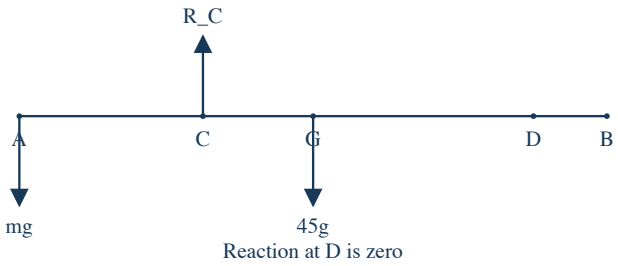


At impending tilt about A: R_B = 0

Key principle

Example 37

A uniform beam AB has mass 45 kg and length 16 m. It rests at C, D, with $AC = 5$ m and $CD = 9$ m. When Woody stands at A, it is on the point of tilting about C. Find his mass.

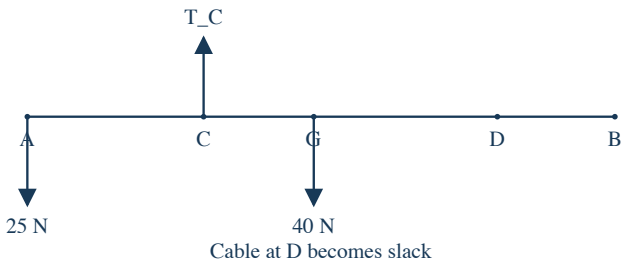


Reaction at D is zero

Working

Suspended System Example 38

A non-uniform rod AB, length 10 m and weight 40 N, hangs from cables at C, D, where $AC = 3$ m and $BD = 2$ m. When a 25 N weight is hung at A, it is on the point of rotating. Find its centre of mass's distance from A.



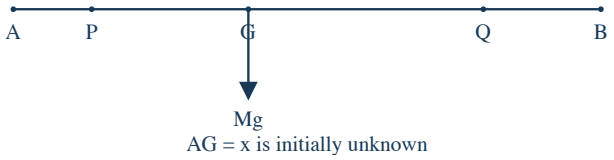
Cable at D becomes slack

Working

Edexcel M1 May 2013, Q6. A non-uniform beam AB, length 15 m, rests at P, Q, where $AP = 2$ m and $QB = 3$ m. A 50 kg child at A brings it to the point of tilting about P; the same child at B brings it to the point of tilting about Q.

a) Find the beam's mass and centre of mass.

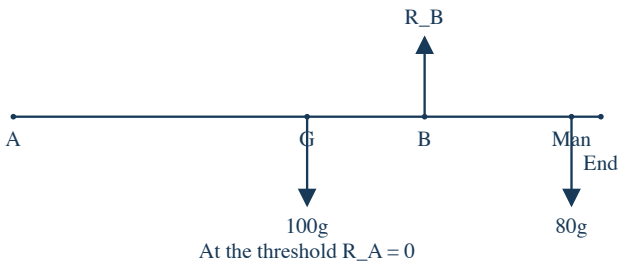
b) The child stands at X, making the reactions equal. Find AX.



a

b

5. A uniform plank has mass 100 kg and length 10 m. It rests on supports at its left end A and at B, 7 m from A. An 80 kg man walks from A towards the other end. How far can he walk past B before tipping begins?

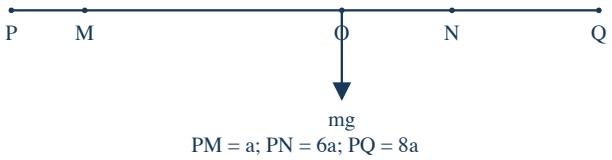


5

6. A non-uniform beam PQ, length $8a$ and mass m, hangs from wires at M, N, with $PM = a, QN = 2a$. A particle of mass $\frac{3}{4}m$ at Q brings it to the point of tipping about N.

a) Show that $ON = \frac{3}{2}a$, where O is the centre of mass.

b) Move the particle to the midpoint. Find the tension at N.

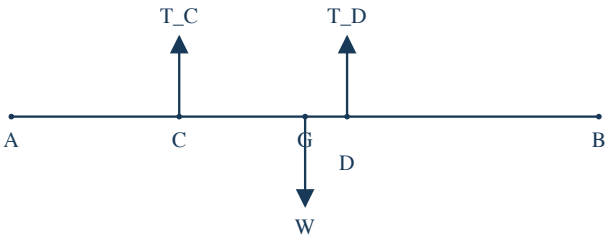


PM = a; PN = 6a; PQ = 8a

a

b

7. A uniform beam AB has length 14 m and weight W. It hangs from vertical cables at C, D, with $AC = 4$ m and $BD = 6$ m. A 180 N weight at A brings it to the point of tilting. Replace it with a weight V at B. Find the maximum V for equilibrium.



First add 180 N at A; then replace by V at B

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